User Instruction Manual for CarbCrete App developed from the paper titled

"Beyond production: concrete carbon emissions benchmarking needs to account for durability performance"

By

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1. General information:

This App, developed in Matlab helps users to calculate the embodied carbon of input concrete mix designs containing supplementary cementitious materials (SCMs). The App considers the following Life cycle assessment (LCA) modules following the modular system boundary of EN 15804 for construction products:

Module A1 – Raw material production

Module A2 – Transportation of raw materials

Module A3 – Concrete production

Module B1 – CO_2 uptake

Module B3 – Concrete repair emission

2. Specific information for each module:

For calculation of Modules A1 to A3, you need to import the concrete mix design information supplied through an excel file. An example template is shown in the file named "coefficients" the alphabets a, b, c, d, e, f, g, h, I, j, k corresponds to water, cement, fly ash, ground granulated blast furnace slag, silica fume, limestone powder, calcined clay, coarse aggregate, fine aggregate, additive and unit weight of concrete respectively. Please maintain this order for your own data. All units are in kg/m^3 of concrete.

After this, you must also import the A1-A3 life cycle impact assessment (LCIA) data. This could be sourced from your any primary data sources (e.g.g, Environmnetal product declaration) or secondary sources (e.g., output from SimaPro or literature). An example template is shown in the file named "distribution_params" containing LCIA data from literatures including average and standard deviations for to water, cement, fly ash, ground granulated blast furnace slag, silica fume, limestone powder, calcined clay, coarse aggregate, fine aggregate, and A3 (concrete manufacturing), while additive value is for min and max of superplasticisers.

Then, information for CO_2 uptake and service life estimation are as described in the original journal paper. CO_2 uptake requires a choice between two modules in the software, BS EN 16757 approach via Lagerblad's constants and a recent model by Vollpracht et al (2024). The service life estimation uses the Vollpracht et al (2024) model.

3. Steps for using CarbCrete

i.) CarbCrete has three main menu tab – "Inputs" "Results" and "Charts"

ii.) Input all required parameters under each of "Material Properties", "Environmental Conditions" and Other Model parameters in the input tab. When using the Vollpracht et al (2024) model for carbonation, you can ignore the following input parameters, carbonation rate, K and the correction factor for K required for SCMs. Note that some inputs have limit restrictions to help avoid ambiguous inputs. Hover over each input box for more details.

iii.) Inputs that are not editable are greyed out.

iv.) Input the number of simulations you want the software to perform for the Monte Carlo uncertainty analysis.

v.) Choose the carbonation model

vi.) Import the "coefficients" and "distribution_params" files one after the other. You will get a pop message to show that files have been successfully imported.

vii.) Run calculation – usually completes in less than 0.1 seconds for the example template import files. An error message will appear if the simulation cannot run due to any issues. Else you get the message of successful simulation completion.

viii.) Go to the results tab to view the results of the simulation. You will find a results tab and a results text summary. You can export the table data to excel by hitting the export results button.

ix.) The chart tab shows you the three plots for the carbon comparison, service life comparison and Cover depth and carbon differences comparison.

If you have any concerns, issues or feedback, contact Donald Nwonu using Donald.chimobi@gmail.com.